

Model Analysis Report

Property Valuation Model

MKM-PV-001

Report Date: 27 March 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Property value by flood risk factor (area=85 m², £8,000/m², all other factors=1.0).

Flood Risk	Factor	Value (£)
Very Low	1.01	686,800.0
Low	0.9900	673,200.0
Medium	0.9500	646,000.0
High	0.8800	598,400.0
Very High	0.8000	544,000.0

Table 2: Property value by age factor (area=85 m², £8,000/m², all other factors=1.0).

Age Band	Factor	Value (£)
New build	1.10	748,000.0
Modern	1.00	680,000.0
Established	0.9500	646,000.0
Period	0.9000	612,000.0
Heritage	0.8500	578,000.0

Table 3: Property value by condition factor (area=85 m², £8,000/m², all other factors=1.0).

Condition	Factor	Value (£)
Excellent	1.15	782,000.0
Good	1.05	714,000.0
Fair	0.9500	646,000.0
Poor	0.8000	544,000.0
Very poor	0.6000	408,000.0

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 4: Stress testing scenarios for Property Valuation Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 5: Performance metrics for Property Valuation Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.